

~~iter 8:~~

Any set containing 1, 3, 1, 4 discard

$\{1, 2, 3\}$ x

$\{1, 2, 4\}$ x

$\{2, 3, 4\}$

22-2-25

Sujit Class 2.

Recap:

AR: $x \rightarrow y$

Supp (AR) = $P(x \cap y) / \text{Total}$

Con (AR) = $\frac{P(y|x)}{P(x)}$

Lift (AR) = $\frac{P(y|x)}{P(y)}$

of AR

Between 2^n & 4^n

Upper bound 3^n

$$\sum_{i=1}^n \binom{n}{i} [2^{n-i} - 1]$$

If Supp & Lift of $x \rightarrow y$ good, Supp & Lift of $y \rightarrow x$ same.

$\{x \cup y\}$ itemset.

Apriori Algo:

Worst case worse than 2^n . Practically runs fast.

If $\text{Supp}(x \cup y) < \text{threshold} \times |D|$

$\Rightarrow x \cup y$ does not occur frequently, we discard all supersets of $x \cup y$.

If an itemset is to be frequent, its subsets should be frequent.

Ex: We call itemset to be freq if $\geq 3 \times$ occurrence

1	1, 2, 3, 4	{1, 3} (3)	{1, 2} (3)	{1, 2, 3} x
2	1, 2, 4	{2, 3} (6)	{1, 3} (1)	{1, 2, 4} x
3	1, 2	{3, 4} (4)	{1, 4} (2)	{2, 3, 4} (2)
4	2, 3, 4	{4} (5)	{2, 3} (3)	{2, 3}
5	2, 3		{2, 4} (4)	{3}
6	3, 4		{3, 4} (3)	No.
7	2, 4			

No more itemsets to generate. STOP.

//_

First list all single elements, check freq. Keep those. Else discard. Then create 2 size sets. check if freq. Else discard. For union of 3 check st. one element is common. check if they have in frequent subsets. Discard those sets as well. Don't combine Disjoint sets.

Pseudocode

T set of all transactions
 ϵ threshold

```

Apriori (T,  $\epsilon$ )
{
  C = {all '1' itemsets}
  while (|C| > 0)
  {
    Find freq of all itemsets in C
    L = sets in C with freq/count  $\geq \epsilon$ 

    C = Apriori Generate (L, k)
  }
}
  
```

Apriori Generate (L, k)

```

{
  for  $x, y \in L$ 
  {
    if  $|x \cup y| = k$ 
    {
      Z =  $x \cup y$ 
    }
  }
}
  
```

if any $k=1$ sized subsets of
 z not in L , do not add z to L .

else add z to L

return L

}

2 if's at same level.

Example:

		Step 1
Step 1	1	$\{1, 2\}$
	2	1, 3
	3	1, 4 x
	4	1, 5 x
	5	2, 3 x
Step 2	6 say infrequent x	2, 4
		2, 5 x
		3, 4
		3, 5
		Step 2 4, 5

Step 2

Step 3	Step 1	Step 3'
1, 2	1 2 3 x	1 2 3
1, 3	1 2 4 x	3, 4 5
2, 4	1 3 4 x	
3, 4	1 3 5 x	
3, 5	2 3 4 x	
4, 5	2 4 5 x	
		3 4 5 Step 2

Output:

Stuff at end of each iteration

$L_1: \{1\} \{2\} \{3\} \{4\} \{5\}$

$L_2: \{1, 2\} \{1, 3\} \{2, 4\} \{3, 4\} \{3, 5\} \{4, 5\}$

$L_3: \{3, 4, 5\}$

To combine sets: $k-1$ set overlap should be there

Say $\{3, 4, 5\}$ & $\{1, 2, 3\}$ left

Can combine? No. Only 3 common.

2 items must be common.

Pseudocode on Wiki

Apriori(T, ϵ)

$L_1 \leftarrow \{ \text{large singleton itemsets} \}$

$k \leftarrow 2$

while L_{k-1} is not empty

$C_k \leftarrow \text{Generate_candidates}(L_{k-1}, k)$

for transactions t in T

$D_t \leftarrow \{ c \text{ in } C_k : c \subseteq t \}$

for candidates c in D_t

count[c] $\leftarrow$ count[c] + 1

$L_k \leftarrow \{ c \text{ in } C_k : \text{count}[c] \geq \epsilon \}$

$k \leftarrow k+1$

1.

return Union $\{ L_k \}$ over all k

//_

Generate - candidates (L, k)

result $\leftarrow$ empty set()

for all $p \in L, q \in L$ where p & q differ
in exactly one element

$c \leftarrow p \cup q$

if $u \in L$ for all $u \subseteq c$ where $|u| = k-1$

result.add(c)

return result.

Different types of association rules
(categorical, hierarchical cyclic)

Eclat Algo:

- Equivalence Class Transformation is a DFS strategy.

FP - Growth

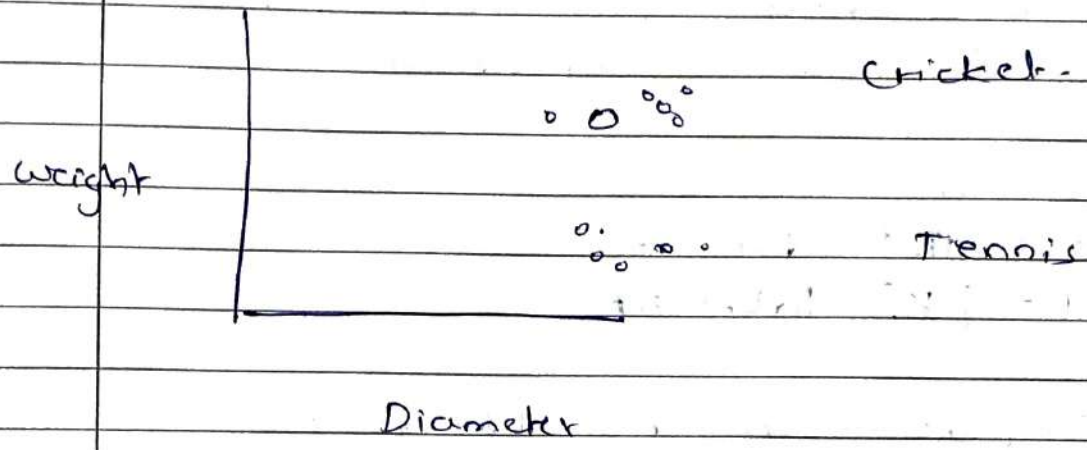
- Frequent pattern growth: A compact data structure called FP tree (Frequent Pattern tree) to compress the dataset.

Book: Data Mining by Vikram Pudi.

Classification:

Classification:

Cricket vs tennis ball based on diameter & weight.



Applications:

Text classification

Classify emails into spam/non-spam

Fraud detection.

We collect different measurements/facts about certain features.

$$x = (x_1, x_2, \dots, x_d)$$

In above eg. x_1, x_2 are dia, weight for one item (one ball)

$$X = (x^1, \dots, x^n)$$

$\subseteq \mathbb{R}^d$ observed

set of all x 's (all balls)

$$Y \in \{1, 2, \dots, K\} = [K]$$

$$y^i \in Y$$

If $K=2$ binary classification, else multi-class classification

So now:

$X \subseteq \mathbb{R}^d$ dataset

$Y = [K]$ label set

Say we have a new data point

x^{new}

y^{new}

$\hat{y}^{\text{new}}$

↑

↑

True label

Predicted label

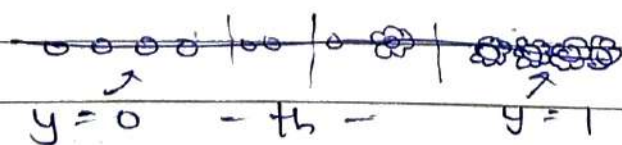
y is target variable.

What is a classifier?

$$f: X \rightarrow Y$$

Focus on binary classification.

Goal to find f .



Here $d=1 \rightarrow$ dimension / feature.

$$f(x) = \begin{cases} 1 & \text{if } x \geq \text{threshold} \\ 0 & \text{otherwise} \end{cases}$$

//_

The threshold value must be chosen, else it will lead to errors in classification.

How to measure 'goodness' of classifier?

How many times you correctly classify.

$$\Pr(\hat{y}^{\text{new}} = y^{\text{new}})$$

predicted = True

Say I have new set of data

$$X^{\text{new}} = \{x^1, \dots, x^l\}$$

$$\# \frac{x^i \text{ correctly classified}}{l}$$

However accuracy is always not a right measure.

If only 1% pop. has cancer, a test of for cancer classifies all people as non-cancer 99% correct. Not useful.

Accuracy:

$$\Pr(\hat{y} = y) = p$$

$y_i \backslash \hat{y}_i$	0	1
0	True Negative	False Positive
1	False Negative	True Positive

Positive Elements, $P = FN + TP$

Negative Elements, $N = TN + FP$

Accuracy : $\frac{TP + TN}{P + N}$

Recall / sensitivity = $\frac{TP}{P} = \frac{TP}{TP + FN}$ (TPR)

Precision = $\frac{TP}{FP + TP}$ (Positive Predictive Value)

FNR : $\frac{FN}{P}$ } $TPR + FNR = 1$

TNR : $\frac{TN}{N}$ } $TNR + FPR = 1$

FPR : $\frac{FP}{N}$

FNR of 1% not good here.

F_1 -Score : Harmonic mean of recall & precision.

$$F_1 = \frac{2}{\frac{1}{\text{precision}} + \frac{1}{\text{recall}}}$$

$$= \frac{2}{\frac{FP+TP}{TP} + \frac{TP+FN}{TP}}$$

$$= \frac{2TP}{2TP + FP + FN}$$